

Complementary Subspaces, Row-Column-spaces

Linear Algebra

Abhishek Sarkar

Department of Mathematics
IIT Jodhpur

Motivation: Complementary Subspaces

- In engineering problems, we often decompose vectors into simpler parts.
- Examples:
 - Signal = useful part + noise
Complex signals are decomposed into orthogonal complements (frequencies or time scales) to filter noise or compress data.
 - Principal Component Analysis (PCA)
Data is decomposed into subspaces corresponding to principal components (maximum variance) and residual subspaces (noise) allowing for dimensionality reduction.
 - Range-Nullspace Decomposition
A system of linear equations $A\mathbf{x} = \mathbf{b}$ is analyzed by decomposing the domain into the row space and null space of matrix A .
 - Force = horizontal component + vertical component
 - State space = controllable part + uncontrollable part
- Complementary subspaces give a mathematical framework for such decompositions.

Why Decomposition?

$$\begin{aligned}(1, 2, 3) &= (1, 0, 0) + (0, 2, 3) \\ &= (1, 2, 0) + (0, 0, 3)\end{aligned}$$

Question

Given a vector $x \in \mathbb{R}^n$, we often ask can we write the following:

$$x = x_1 + x_2 \quad ?$$

- Can we split x into two meaningful components?
- Is the decomposition unique?
- What properties do x_1 and x_2 have?
- This leads to the idea of **complementary subspaces**.

Definition: Complementary Subspaces

$$X = Y + Z$$

- Let X be a vector space and let $Y, Z \subset X$ be subspaces.

Definition (Complementary Subspaces)

Y and Z are called **complementary subspaces** if

$$X = Y \oplus Z,$$

↖ direct sum.

that is:

- Every $x \in X$ can be written as $x = y + z$
- The decomposition is **unique**

}

Definition: Complementary Subspaces

- Let X be a vector space and let $Y, Z \subset X$ be subspaces.

Definition (Complementary Subspaces)

Y and Z are called **complementary subspaces** if

$$X = Y \oplus Z,$$

that is:

- Every $x \in X$ can be written as $x = y + z$
- The decomposition is **unique**

Uniqueness means:

$$Y \cap Z = \{0\}.$$

What Does $X = Y \oplus Z$ Mean?

$$X = Y \oplus Z$$

means:

- $X = Y + Z$ (together they span X)
- $Y \cap Z = \{0\}$ (no overlap except zero)

What Does $X = Y \oplus Z$ Mean?

$$X = Y \oplus Z$$

means:

- $X = Y + Z$ (together they span X)
- $Y \cap Z = \{0\}$ (no overlap except zero)

Key idea:

Every vector splits into two independent parts.

Geometric Example in $\mathbb{R}^2$

Example

Let us consider two subspaces of $\mathbb{R}^2$:

$$Y = \text{span}\{(1, 0)\}, \quad Z = \text{span}\{(0, 1)\}.$$

$\rightarrow$ x axis

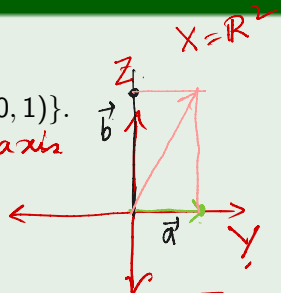
$\rightarrow$ y axis

- Y is the x-axis and Z is the y-axis

- $\mathbb{R}^2 = Y \oplus Z$

Every vector (a, b) can be written as:

$$(a, b) = (a, 0) + (0, b).$$



$$X = Y + Z$$

Qn: $Y \cap Z = \{(0, 0)\}$

$$X = Y \oplus Z$$

$Y = xy$ - plane

$Z = yz$ - plane

$$X = \mathbb{R}^3 = Y + Z$$

$$Y \cap Z \neq \{(0,0,0)\}$$

Not a direct sum.

Non-example
of direct sum.

$$\rightarrow \{(a,b,0) : a,b \in \mathbb{R}\}$$

$$\rightarrow \{(0,c,d) : c,d \in \mathbb{R}\}$$

$$(1,2,3) = \underbrace{(1,2,0)}_Y + \underbrace{(0,0,3)}_Z = \underbrace{(1,0,0)}_Y + \underbrace{(0,2,3)}_Z$$

Equivalent Definitions

$X = Y \oplus Z$ is a direct sum then the following are equivalent:

Equivalent Definitions

- $Y \cap Z = \{0\}$.
- Two non-zero vectors y and z are linearly independent whenever $y \in Y$ and $z \in Z$.
- The only one way to choose $y \in Y$ and $z \in Z$ such that $y + z = 0$ is to choose $y = z = 0$.
- For any $x \in Y + Z$, there exists a unique couple $y \in Y$ and $z \in Z$ such that $x = y + z$.

① $\Rightarrow$ ④ let $Y \cap Z = \{0\}$

we have to prove ④

Let the decomposition is not unique!

(Method of contradiction)

Suppose $x = y_1 + z_1 = y_2 + z_2$ $y_1, y_2 \in Y$

$$\Rightarrow y_1 - y_2 = -z_1 + z_2 = -(z_1 - z_2) \quad z_1, z_2 \in Z.$$

Since Y & Z both are subspaces

$$y_1 - y_2 \in Y \quad \text{and} \quad y_1 - y_2 \in Z. \quad (\text{since } z_1 - z_2 \in Z)$$

$$\Rightarrow y_1 - y_2 \in Y \cap Z \Rightarrow y_1 - y_2 = 0 \Rightarrow y_1 = y_2$$

$$\text{Similarly } z_1 - z_2 \in Z, \quad z_1 - z_2 \in Y \Rightarrow z_1 - z_2 \in Z \cap Y = \{0\} \\ \Rightarrow z_1 = z_2.$$

Non-Example

Non-Example

Let us consider two subspaces of $\mathbb{R}^2$:

$$Y = \text{span}\{(1, 0)\}, \quad Z = \text{span}\{(2, 0)\}.$$

- Both lie on the x-axis
- $Y \cap Z \neq \{0\}$
- Decomposition is not unique

Conclusion: These are NOT complementary.

Non-Example

Non-Example

Let us consider two subspaces of $X = \mathbb{R}^3$:



$$X_1 = xy\text{-plane}, \quad X_2 = yz\text{-plane}.$$

- An element $(1, 2, 3) \in \mathbb{R}^3$ can be written as

$$(1, 2, 3) = (1, 2, 0) + (0, 0, 3) = (1, 0, 0) + (0, 2, 3) = (1, 1, 0) + (0, 1, 3).$$

- $X_1 \cap X_2 \neq \{(0, 0, 0)\}$
- Decomposition is not unique

Conclusion: These are NOT complementary. Although, $X_1 + X_2 = \mathbb{R}^3$.

Orthogonal Complement (Important Case)

- In $\mathbb{R}^n$ with dot product:

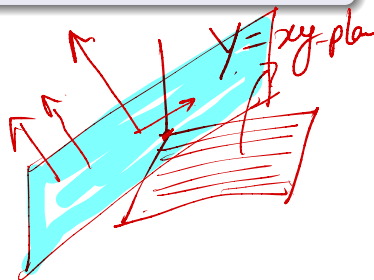
Orthogonal Complement

For a subspace Y , define

$$Y^\perp = \{x : x \cdot y = 0 \text{ for all } y \in Y\}.$$

$Y^\perp$ -perp

$$\mathbb{R}^n = Y \oplus Y^\perp$$



Orthogonal Complement (Important Case)

- In $\mathbb{R}^n$ with dot product:

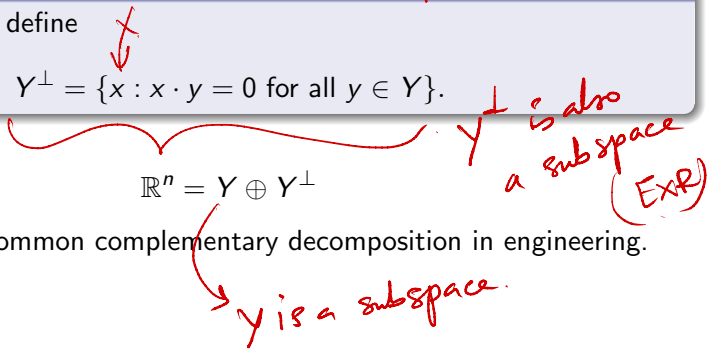
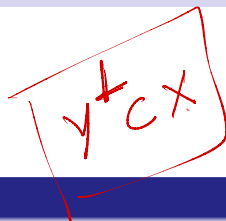
Orthogonal Complement

For a subspace Y , define

$$Y^\perp = \{x : x \cdot y = 0 \text{ for all } y \in Y\}.$$

$$\mathbb{R}^n = Y \oplus Y^\perp$$

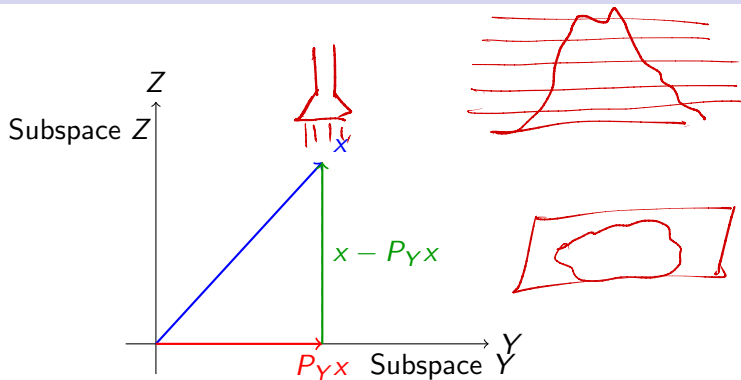
This is the most common complementary decomposition in engineering.



Engineering Interpretation

- Signal processing: signal + noise
- Mechanics: parallel + perpendicular forces
- Control systems: controllable + uncontrollable subspaces
- Numerical methods: solution space + error space

Projection onto a Subspace

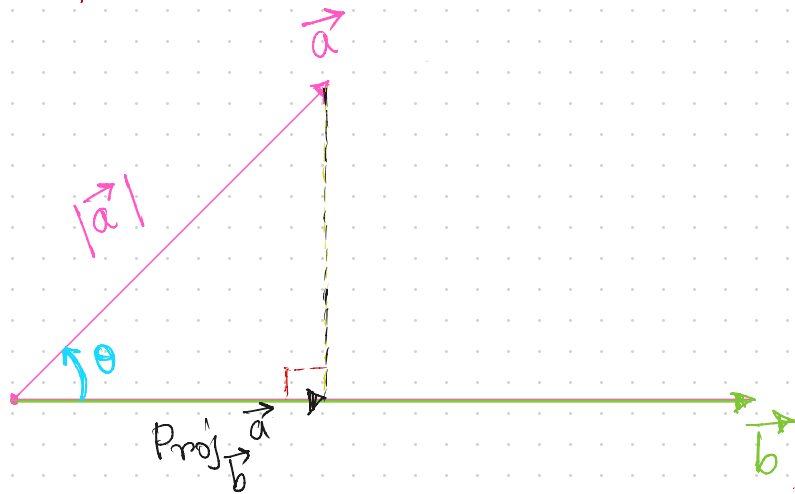


$$x = P_Y x + (x - P_Y x), \quad \mathbb{R}^2 = Y \oplus Z$$

- The projection of $\vec{a}$ on $\vec{b}$ (written as $\text{proj}_{\vec{b}} \vec{a}$)

$$(2,0) = \vec{b}$$

$$\vec{a} = (1,3)$$



$$\text{Proj}_{(2,0)} (1,3)$$

$$= \frac{(1,3) \cdot (2,0)}{|(2,0)|^2} (2,0)$$

$$= \frac{2}{4} (2,0)$$

$$= \frac{1}{2} (2,0)$$

$$= (1,0)$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} \Rightarrow \text{adjacent} = |\vec{a}| \cos \theta$$

Therefore,

$$\text{Proj}_{\vec{b}} \vec{a} = |\vec{a}| \cos \theta \frac{\vec{b}}{|\vec{b}|} = |\vec{a}| \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \frac{\vec{b}}{|\vec{b}|}$$

$$= \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b} \quad (|\vec{a}| \neq 0)$$

Projection Viewpoint

If $X = Y \oplus Z$, then there exists a **projection**

$$P : X \rightarrow Y$$

such that:

- $P^2 = P$
- $\ker P = Z$

Projection Viewpoint

If $X = Y \oplus Z$, then there exists a **projection**

$$P : X \rightarrow Y$$

such that:

- $P^2 = P$
- $\ker P = Z$

Every vector x satisfies:

$$x = Px + (x - Px).$$

Summary

- Complementary subspaces allow a unique decomposition of vectors
- They generalize the idea of perpendicular directions
- Orthogonal complements are the most important case
- Widely used in engineering and applied sciences

Thank You

Questions?